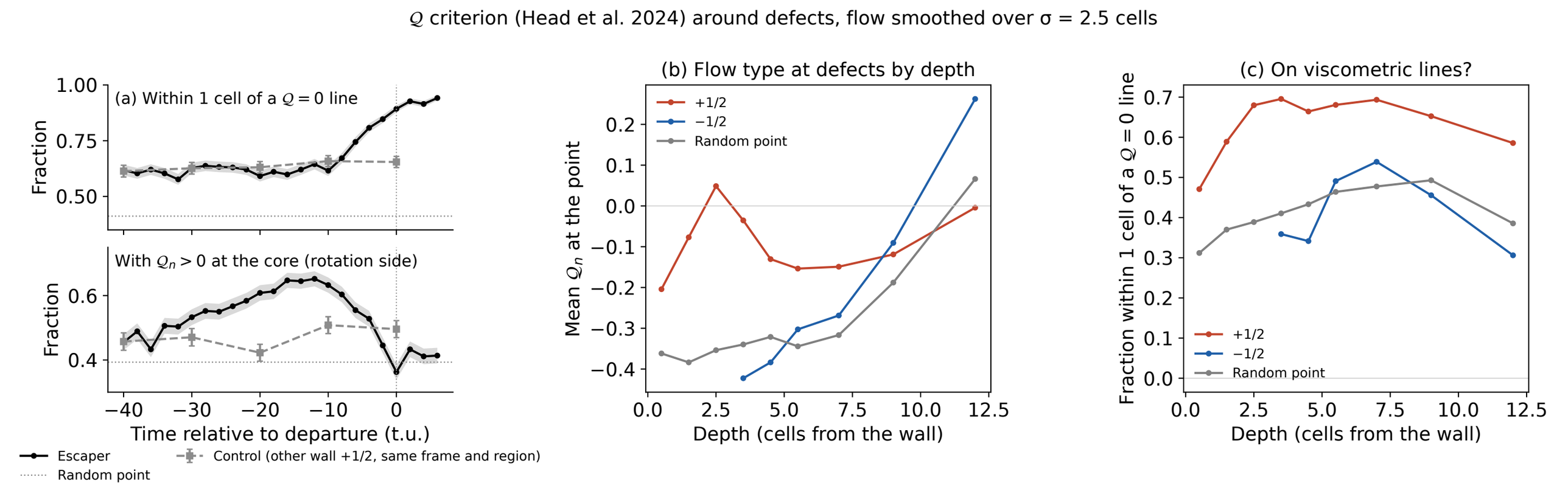
$\mathcal{Q}$ criterion (Head et al. 2024) around defects, flow smoothed over σ = 2.5 cells
(a) Within 1 cell of a $\mathcal{Q} = 0$ line
Fraction
With $\mathcal{Q}_n > 0$ at the core (rotation side)
Time relative to departure (t.u.)
Escaper
Control (other wall +1/2, same frame and region)
Random point
(b) Flow type at defects by depth
Mean $\mathcal{Q}_n$ at the point
+1/2
−1/2
Random point
Depth (cells from the wall)
(c) On viscometric lines?
Fraction within 1 cell of a $\mathcal{Q} = 0$ line
Depth (cells from the wall)